

Whole-Body Impedance Control of a Humanoid Robot Based on Human-Human Demonstration for Human-Robot Collaboration

Chenzui Li¹, Junjia Liu¹, Tao Teng^{1,2}, Shixiong Wang^{1,2}, Sylvain Calinon³, and Fei Chen^{†1,2}

Abstract—This paper proposes a novel whole-body impedance control method for the Collaborative dual-arm Robot manipulator (CURI) in Human-Robot Collaboration (HRC). The method enables CURI to adapt its physical behavior to human motion while following trajectories learned from human-human demonstrations. A whole-body impedance controller coordinates the robot joints to achieve desired Cartesian space impedance. Collaborative tasks are captured from human-human demonstrations and represented using a Task-parameterized Gaussian Mixture Model (TP-GMM). Electromyography (EMG) sensors record muscle activities to estimate human impedance profiles, which are then mimicked by a variable impedance controller. An adaptive parameter is introduced to adjust robot stiffness based on spatial displacement between the robot and human, ensuring safe and efficient interaction. Experimental validation through confrontational Tai Chi pulling/pushing tasks demonstrates the superiority of the proposed adaptive impedance method over the fixed impedance controller.

I. INTRODUCTION

Collaborative robots (cobots) have emerged as a transformative solution for enhancing human-robot interactions in both industrial and domestic settings. While fully robotic systems often lack the flexibility and adaptability required for complex tasks, human-only operations can lead to fatigue, errors, and increased labor costs [1]. Cobots bridge this gap by enabling co-manipulation, where humans guide operations while robots adapt to human behaviors, reducing physical strain and improving efficiency. For instance, cobots excel in tasks like handling bulky objects, where humans act as leaders while robots provide physical support and adapt to human motions [2], [3]. Collaborative sawing is another example, where robots dynamically adjust their behavior based on human fatigue levels, ensuring smooth task execution [4]. Despite these advancements, significant challenges remain in designing robot morphology for complex tasks, developing adaptive control strategies, and enabling cobots to recognize and comprehend human behaviors [5], [6], [7].

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¹Chenzui Li, Junjia Liu, Tao Teng, Shixiong Wang, and Fei Chen are with the Department of Mechanical and Automation Engineering, T-Stone Robotics Institute, The Chinese University of Hong Kong, Hong Kong (e-mail: czli@mae.cuhk.edu.hk, jiliu@mae.cuhk.edu.hk, tao.teng@ieee.org, sxwang@hkclrl.hk, f.chen@ieee.org).

²Tao Teng, Shixiong Wang, and Fei Chen are also with the Hong Kong Center for Logistics Robotics, Hong Kong.

³Sylvain Calinon is with the Idiap Research Institute, Martigny, Switzerland (e-mail: sylvain.calinon@idiap.ch).

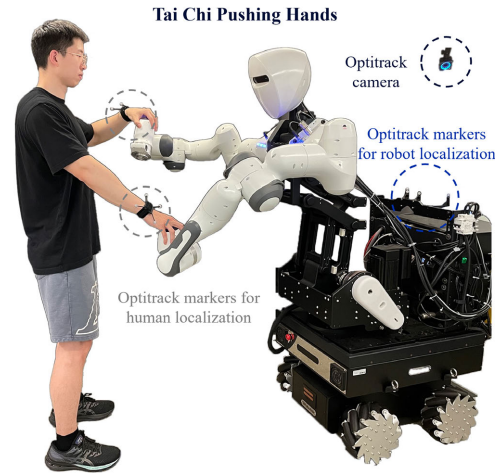


Fig. 1. Illustration of human-robot collaborative Tai Chi pushing hand.

In this work, we introduce a novel mobile humanoid robot, CURI, designed to address these challenges in human-robot collaboration (HRC). A key challenge in mobile humanoid robots is the design of whole-body impedance controllers, which must balance multiple objectives such as stability, safety, and task execution. Traditional approaches often focus on task space impedance control, which simplifies planning in low-dimensional spaces while utilizing redundant degrees of freedom (DOFs) for additional functionalities like collision avoidance. Inspired by prior work [8], [9], we propose a novel whole-body impedance controller for CURI, enabling intuitive task execution while leveraging redundant DOFs for adaptive control.

Effective interactive control strategies are essential for safe and efficient HRC. Impedance control, which maps force and position, is particularly critical for physical human-robot interaction (pHRI). While variable impedance controllers (VICs) have been proposed to modulate stiffness and damping, many existing approaches lack guaranteed stability during interaction. For example, some methods adjust stiffness based on motion variance and speed but fail to ensure consistent interaction quality [2]. Others rely on electromyography (EMG) signals to transfer human stiffness to robots, but signal instability can lead to unsafe robot behavior [4]. To address these limitations, we propose a novel VIC design that dynamically adjusts both stiffness and damping while ensuring stability, leveraging the robot's redundant DOFs for adaptive control.

Learning from Demonstration (LfD) is a widely adopted approach for modeling human behaviors in HRC. Tech-

niques like Task-parameterized Gaussian Mixture Models (TP-GMM) enable robots to learn and adapt complex human motions to new scenarios [10]. By integrating EMG signals, robots can infer human impedance profiles and adapt their behavior accordingly, as demonstrated in tasks like collaborative sawing [4]. However, most studies focus on single-human or robot-guided demonstrations, leaving human-human collaborative demonstrations largely unexplored. This work leverages human-human demonstrations to extract rich kinematic and dynamic information, enabling CURI to perform collaborative tasks more effectively.

To address these challenges, we present the following contributions:

- 1) A whole-body impedance controller for CURI, coordinating its arms, torso, and mobile chassis to generate desired stiffness and damping in Cartesian space.
- 2) An adaptive impedance controller that adjusts robot stiffness and damping based on real-time spatial distance between humans and robots, ensuring safe and reliable interaction.
- 3) A TP-GMM-based approach to represent and adapt human demonstrations for new tasks, enabling CURI to generate appropriate movements in dynamic scenarios.

II. WHOLE-BODY IMPEDANCE CONTROLLER

This section introduces the whole-body impedance controller for CURI, designed to ensure safe physical human-robot collaboration. CURI is a dual-arm mobile platform (Fig. 1) with two Tool Center Points (TCPs), each corresponding to one arm. The TCPs are analyzed separately for whole-body dynamics. The linear and angular velocity of a TCP relative to the world frame is the sum of velocities generated by the manipulators, torso, and mobile base. The whole-body Jacobian $J_{wb} \in \mathbb{R}^{6 \times 13}$ for one TCP is formulated as:

$$\dot{\mathbf{x}}_{TCP} = J_{wb} \dot{\mathbf{q}} \quad (1)$$

where $\dot{\mathbf{x}}_{TCP} \in \mathbb{R}^6$ represents the linear and angular velocities of the TCP frame Σ_{TCP} w.r.t. the world frame Σ_W . $J_{wb} = (J_{base} \ J_{torso} \ J_{arm}) \in \mathbb{R}^{6 \times 13}$, where $J_{base} \in \mathbb{R}^{6 \times 3}$, $J_{torso} \in \mathbb{R}^{6 \times 3}$ and $J_{arm} \in \mathbb{R}^{6 \times 7}$ represent the contribution of the motion of the base, torso and arm. $\dot{\mathbf{q}} = (\dot{\mathbf{q}}_{base}^T \ \dot{\mathbf{q}}_{torso}^T \ \dot{\mathbf{q}}_{arm}^T)^T \in \mathbb{R}^n$ represents whole-body joint velocity vector, where $\dot{\mathbf{q}}_{base} \in \mathbb{R}^3$, $\dot{\mathbf{q}}_{torso} \in \mathbb{R}^3$ and $\dot{\mathbf{q}}_{arm} \in \mathbb{R}^7$ respectively represent the joint velocity vectors of the base, torso, and arm.

Since both the self-designed torso and the mobile base (Robotnik SUMMIT-XL STEEL) utilize velocity-based control with high-gain low-level controllers, their dynamics can be omitted. This simplification allows us to ignore external dynamic effects from the manipulator and coupling dynamics between the torso and mobile base. To implement a whole-body impedance control law, a force-torque interface is preferred. Therefore, we designed a Cartesian admittance controller based on the velocity interface for both the torso and mobile base. The dynamics of CURI can be expressed as follows:

$$\begin{pmatrix} M_b & 0 & 0 \\ 0 & M_t & 0 \\ 0 & 0 & M_a \end{pmatrix} \begin{pmatrix} \ddot{\mathbf{q}}_{base} \\ \ddot{\mathbf{q}}_{torso} \\ \ddot{\mathbf{q}}_{arm} \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \mathbf{G}_a \end{pmatrix} + \begin{pmatrix} D_b & 0 & 0 \\ 0 & D_t & 0 \\ 0 & 0 & C_a \end{pmatrix} \begin{pmatrix} \dot{\mathbf{q}}_{base} \\ \dot{\mathbf{q}}_{torso} \\ \dot{\mathbf{q}}_{arm} \end{pmatrix} = \begin{pmatrix} \tau_b \\ \tau_t \\ \tau_a \end{pmatrix} + \begin{pmatrix} \tau_b^{ext} \\ \tau_t^{ext} \\ \tau_a^{ext} \end{pmatrix} \quad (2)$$

where $M_b \in \mathbb{R}^{3 \times 3}$ and $D_b \in \mathbb{R}^{3 \times 3}$ are respectively the virtual inertial and virtual damping set for the base in the admittance controller. While $M_t \in \mathbb{R}^{3 \times 3}$ and $D_t \in \mathbb{R}^{3 \times 3}$ are respectively the virtual inertial and virtual damping set for the torso in the admittance controller. $\tau_b \in \mathbb{R}^3$ and $\tau_b^{ext} \in \mathbb{R}^3$ are respectively the commanded and the external torques for the base. While $\tau_t \in \mathbb{R}^3$ and $\tau_t^{ext} \in \mathbb{R}^3$ are respectively the commanded and the external torques for the torso. $M_a \in \mathbb{R}^{7 \times 7}$, $C_a \in \mathbb{R}^7$ and $G_a \in \mathbb{R}^7$ are the inertia matrix, gravity term, Coriolis and centrifugal terms, respectively. $\tau_a \in \mathbb{R}^7$ and $\tau_a^{ext} \in \mathbb{R}^7$ are respectively the commanded and the external torques for the robot arm.

Considering the movement of robot's end-effector as a single unit mass moving in Cartesian space under the effect of a control input τ_c and interactive torque τ_{ext} . Hence, the Cartesian-space impedance model to dynamically connect the robot positions, velocities and accelerations with the interactive forces is considered as:

$$M_x \ddot{\bar{\mathbf{x}}} + D(t) \dot{\bar{\mathbf{x}}} + K(t) \bar{\mathbf{x}} = \tau_{ext} \quad (3)$$

where $\bar{\mathbf{x}} = \mathbf{x}_d - \mathbf{x} \in \mathbb{R}^m$ in which $\mathbf{x}_d$ is the pose of an attractor point and $\mathbf{x}$ the robot end effector pose, and $M_x \in \mathbb{R}^{m \times m}$, $D(t) \in \mathbb{R}^{m \times m}$, $K(t) \in \mathbb{R}^{m \times m}$ are the apparent virtual mass, desired damping, and desired stiffness matrices, respectively. It is noted that M_x is defined as constant and $D(t)$, $K(t)$ time-varying in the proposed controller.

Finally, with the given robot dynamic (2) and impedance model (3), the cartesian impedance controller proposed in [9] can be extended to CURI in a straightforward manner, while its stability can also be guaranteed. Then the whole-body impedance controller for CURI can be expressed as:

$$(\tau_b^T \ \tau_t^T \ \tau_a^T)^T = \tau_{task} + \tau_{comp} + \tau_{null} \quad (4)$$

The τ_{task} corresponding to the given main task is:

$$\tau_{task} = J_{wb}^T (-K(t) \bar{\mathbf{x}} - D(t) \dot{\bar{\mathbf{x}}}) \quad (5)$$

The τ_{comp} is designed to compensate the nonlinear terms of the system, and can be designed as:

$$\tau_{comp} = \begin{pmatrix} 0 \\ 0 \\ M_a \ddot{\mathbf{q}}_{arm} + C_a \dot{\mathbf{q}}_{arm} + G_a \end{pmatrix} \quad (6)$$

and the null-space task input is defined as:

$$\tau_{null} = N(\mathbf{q})(\tau_{subtask}) \quad (7)$$

where the $\tau_{subtask}$ represents the corresponding torque for the given sub task and $N(\mathbf{q}) \in \mathbb{R}^{13 \times 13}$ is the projection matrix in order to prevent interference with the cartesian

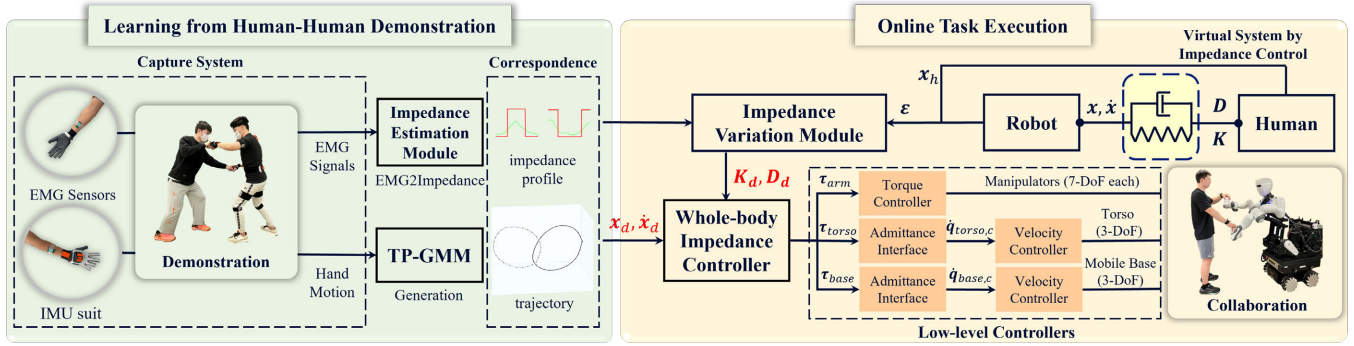


Fig. 2. The overall framework scheme from human-human demonstration to human-robot collaboration/interaction. Humans are able to teach CURI the collaborative/interactive tasks by recording multi-modal data such as the IMU signals, global 3D position/rotation data, EMG signals, etc., and generating the Cartesian trajectory profiles with TP-GMM as well as the impedance profiles through EMG-based Impedance Estimation Module offline. Then, the estimated impedance profiles are sent to construct an Impedance Variation Module for robot impedance generation based on the relative distance between the human and the robot online. Finally, the desired trajectories and impedance are sent to CURI whole-body impedance controller for task execution.

impedance behavior of main task. In this paper, the following projection in [11] is employed:

$$N(q) = I - J_{wb}^T J_{wb}^{+T}. \quad (8)$$

where the J_{wb}^+ expresses the generalized inverse of J_{wb} .

Finally, after obtaining the designed torques $\tau_{wb} = (\tau_b^T, \tau_t^T, \tau_a^T)^T$, the τ_a is sent to the robot arm directly through its torque command interface. But the mobile base and torso received velocity command only, and the τ_b and τ_t can respectively generate its corresponding velocity command $\dot{q}_{b,c}$ and $\dot{q}_{t,c}$ through the corresponding admittance interfaces in (2).

The above analysis provides a controller for one TCP of CURI. However, applying the same controller to the second TCP may cause conflicts in commands sent to the base and torso. To address this, we first design a Cartesian impedance controller for one TCP (referred to as the *follower-arm*). The null-space torque $\tau_{subtask}$ in (7) for the *follower-arm* can be used for user-defined subtasks, such as optimizing joint configurations. The calculated base torque $\tau_{b,f}$ and torso torque $\tau_{t,f}$ for the *follower-arm* are treated as subtasks for the second TCP (referred to as the *leader-arm*), while the arm torque τ_{arm} is directly sent to the *follower-arm*. The subtask torque for the *leader-arm* is defined as:

$$\tau_{subtask} = (\tau_{b,f}^T, \tau_{t,f}^T, \mathbf{0}_{1 \times 7}) \quad (9)$$

This approach prevents command conflicts between the two TCPs and ensures whole-body compliance. The resulting control framework integrates all four parts of CURI, as shown in Fig. 1.

The impedance model matrices (stiffness K and damping D) are assumed diagonal, with $i \in [1, \dots, m]$ DoFs. The stiffness and damping ranges are defined by maximum (K_{max} , D_{max}) and minimum (K_{min} , D_{min}) values. The system impedance is adjusted as:

$$\begin{aligned} K(t) &= K_{min} + S_K(t)(K_{max} - K_{min}) \\ D(t) &= D_{min} + S_D(t)(D_{max} - D_{min}) \end{aligned} \quad (10)$$

Here, S_K and S_D are diagonal matrices with differentiable gains $\delta_i \in [0, 1]$.

This impedance model enables dynamic adjustment of the robot's physical behavior by varying stiffness and damping while tracking attractor points. Based on this, we propose a variable impedance controller (VIC) to generate $K(t)$ and $D(t)$ for task-specific requirements.

III. TASK DEFINITION

Tai Chi pulling/pushing hands, a traditional Chinese martial art, involves unarmed confrontation using upper limbs, torso, and lower limbs. It is widely practiced for its health benefits, particularly among elders [12]. This activity can be categorized into fixed-step and moving-step techniques, with the former serving as the foundation. Fixed-step techniques include pushing motions in flat circles, vertical circles, and figure-eight patterns [13]. In this work, we focus on fixed-step unimanual pulling along a straight line and bimanual pushing in flat and vertical circles for human-robot interaction, as illustrated in Fig. 3.

During Tai Chi pulling/pushing hands, the physical behaviors of two individuals are reciprocal. Specifically, when performing pulling or pushing motions, one person exhibits stiff behavior while the other remains compliant during the first half of the trajectory. These roles reverse in the second half, with the initially stiff person becoming compliant and vice versa. To replicate this in human-robot collaboration, the robot must generate reciprocal physical behavior relative to the human opponent at each stage. We divide the circular motion of each hand into two phases: 'Yin' (softness) and 'Yang' (rigidity), inspired by traditional Chinese philosophy [14]. During the 'Yin' phase, the robot exhibits low impedance to comply with human motion, while during the 'Yang' phase, the robot end effectors become stiff to pull or push the human hands.

IV. METHODOLOGY

In this section, the stiffness/damping variables and the reference trajectories of the Cartesian-space impedance model will be generated by our proposed method for CURI end effectors to perform appropriate physical behavior while

interacting with humans. An EMG-based impedance estimation method is given to construct the human end effectors' impedance profiles during human-human demonstration. Then, a variable impedance controller is proposed to generate desired impedance of CURI robot based on the generated impedance profiles and the adaptive parameters according to real-time distance feedback. Finally, the TP-GMM method is proposed for learning motion representation from the demonstrations. The schematic diagram of the system is shown in Fig. 2.

A. EMG-based Impedance Estimation Method

As defined in Section III, there are two phases of the collaborative Tai Chi pulling/pushing hands including the 'Yin' phase and the 'Yang' phase from the robot perspective. Assume that the robot is able to distinguish the two phases of its motion as a priori if the reference trajectory is given based on the demonstrations. Then the robot is able to be compliant in the 'Yin' phase and is expected to become relatively stiff in the 'Yang' phase compared to the human opponent. In this part, we propose to generate proper impedance for the robot by learning the impedance profile of the demonstrator when presenting Tai Chi with the other.

Assume that the EMG signals from the demonstrator are defined as $\mathbf{u} = [u_1, u_2, \dots, u_k] \in \mathbb{R}^k$ with k EMG sensors in measurement at each time step. They are further processed by filters and normalised using Maximal Voluntary Contraction (MVC). Then, the muscular activation levels based on the MVC and the processed EMG data can be calculated as $\mathbf{A} = [A_1, A_2, \dots, A_k] \in \mathbb{R}^k$. The estimated human stiffness model [15] based on the muscular activation levels of the same muscle group has been modified to include multi-muscle groups and can be further formulated as:

$$K_h = \lambda_1 \frac{1 - e^{-\lambda_2(\omega_1 A_1 + \omega_2 A_2 + \dots + \omega_k A_k)}}{1 + e^{-\lambda_2(\omega_1 A_1 + \omega_2 A_2 + \dots + \omega_k A_k)}} \quad (11)$$

where K_h represents the estimation of human impedance trend, Parameter λ_1 defines the maximum amplitude of mapping and λ_2 defines the shape of mapping. $\boldsymbol{\omega} = [\omega_1, \omega_2, \dots, \omega_k] \in \mathbb{R}^k$ represents the weights of each part of the muscle contributing to impedance generation. Finally, a threshold value defined as λ_3 is customized to determine whether the human impedance is relatively high ('Yang' phase of Tai Chi) or relatively low ('Yin' phase of Tai Chi). For example, the red line in the bottom figure of Fig. 5 illustrates the switching between 'Yin' and 'Yang' of one demonstrator during Tai Chi pushing hands.

B. Variable Impedance Controller with Adaptive parameters

As defined in Section II, the change in the robot stiffness (respectively damping) is defined to vary between values K_1 and K_2 (respectively D_1 and D_2) in this task. Hence, a variable impedance controller is designed to achieve the learned physical behavior from IV-A:

$$\begin{aligned} \mathbf{K}(t) &= K_1 + \mathbf{S}_K(t)(K_2 - K_1) \\ \mathbf{D}(t) &= D_1 + \mathbf{S}_D(t)(D_2 - D_1) \end{aligned} \quad (12)$$

In order to avoid instability of the impedance control method due to arbitrary variations in impedance parameters, some passivity conditions of the varying impedance parameters should be satisfied to guarantee system stability [16], [17]. The constraint for our task is chosen as follows:

$$\begin{aligned} 0 &\leq \mathbf{K}(t) - \dot{\mathbf{D}}(t) \\ \Rightarrow 0 &\leq \mathbf{S}_K(t) - \dot{\mathbf{S}}_D(t) \end{aligned} \quad (13)$$

then the system stability will be ensured if we satisfy the condition while designing the variable impedance controller.

In human-robot collaboration scenarios, a predefined variable impedance controller for the robot which imitates the physical behavior of the human in the demonstration is not enough to be applied. It will make the robot dangerous to humans when interacting with each other due to the lack of online feedback control parameters. In this context, an adaptive parameter ϵ based on the spatial displacement relationship between the robot and the human opponent during the collaboration is proposed to vary the robot stiffness in a proper way during the interaction.

$$\epsilon = \begin{cases} 0, & \|\mathbf{x}_h - \mathbf{x}\| \leq d \\ \frac{\|\mathbf{x}_h - \mathbf{x}\| - d}{\|\mathbf{x}_h - \mathbf{x}\|}, & \|\mathbf{x}_h - \mathbf{x}\| > d \end{cases} \quad (14)$$

where $\mathbf{x}_h$ is the human hand pose and $\mathbf{x}$ the robot end effector pose in Cartesian space, d represents the predefined safe spatial distance between the human hand and the robot end effector. By adding the adaptive parameter to the customized variable impedance controller, the variety of robot end effector's stiffness is related to the spatial distance between human and robot. Specifically, the smaller the distance between the two agents, the lower the robot stiffness. Besides, the target stiffness is set to be a constant if they are judged to be in an interaction area, where their distance is less than d . Note that we only consider the stiffness variation to adapt the changes in relative position.

Hence, the diagonal matrices $\mathbf{S}_K$ and $\mathbf{S}_D$ are expressed as linear, discrete-time models which have the forms:

$$\mathbf{S}_K(t) = \begin{cases} \alpha \epsilon \mathbf{I} + (1 - \alpha) \mathbf{S}_K(t-1), & \text{Yin} \\ \alpha \gamma (1 + \epsilon) \mathbf{I} + (1 - \alpha) \mathbf{S}_K(t-1), & \text{Yang} \end{cases} \quad (15a)$$

$$\mathbf{S}_D(t) = \begin{cases} (1 - \beta) \mathbf{S}_D(t-1), & \text{Yin} \\ \beta \gamma \mathbf{I} + (1 - \beta) \mathbf{S}_D(t-1), & \text{Yang} \end{cases} \quad (15b)$$

where $\mathbf{I} \in \mathbb{R}^{m \times m}$ is a unit diagonal matrix. α and β are the tuning parameters to prevent sudden changes in value while γ is a scale index.

By substituting the (15a) and (15b) into the stability constraint (13), we have:

$$\begin{aligned} \mathbf{S}_K(t) - \dot{\mathbf{S}}_D(t) &= \mathbf{S}_K(t) - \mathbf{S}_D(t) + \mathbf{S}_D(t-1) \\ &= \mathbf{S}_K(t) + \beta \mathbf{S}_D(t-1) \geq 0 \end{aligned} \quad (16a)$$

$$\begin{aligned} \mathbf{S}_K(t) - \dot{\mathbf{S}}_D(t) &= [\alpha(1 + \epsilon) - \beta] \gamma \mathbf{I} + \\ &\quad (1 - \alpha) \mathbf{S}_K(t-1) + \beta \mathbf{S}_D(t-1) \geq 0 \end{aligned} \quad (16b)$$

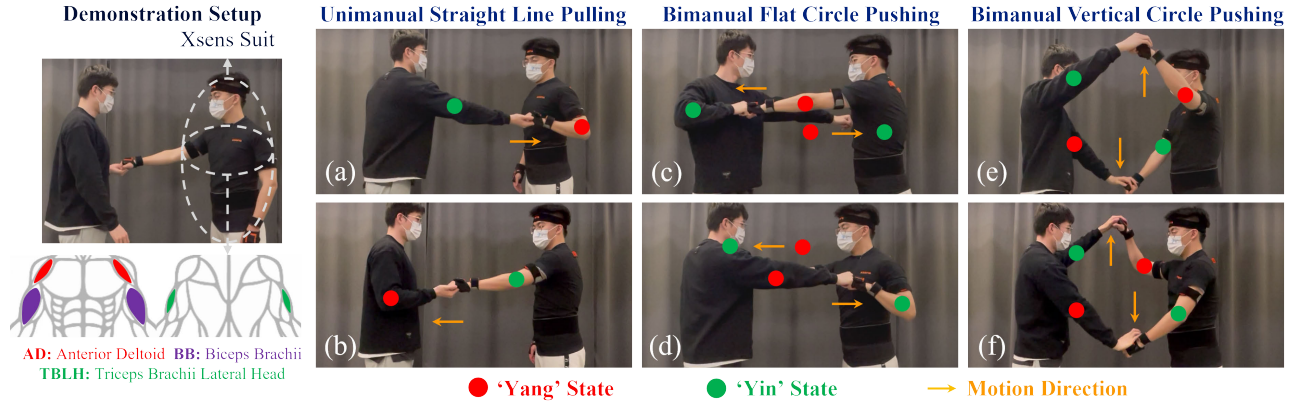


Fig. 3. Human-human demonstrations were conducted to capture motion data for CURI. One demonstrator wore an Xsens suit, while EMG sensors were attached to several upper-body muscles to record muscle activity. Three fixed-step Tai Chi pulling/pushing hand motions were demonstrated: unimanual straight-line pulling (repetition of $a, \rightarrow, b, \rightarrow, a$), bimanual flat-circle pushing (repetition of $c, \rightarrow, d, \rightarrow, c$), and bimanual vertical-circle pushing (repetition of $e, \rightarrow, f, \rightarrow, e$). Each motion cycle was divided into ‘Yin’ (softness) and ‘Yang’ (rigidity) phases, determined using an EMG-based impedance estimation method. Red solid circles indicate hands in the ‘Yang’ phase, while green solid circles represent the ‘Yin’ phase. Orange arrows denote the direction of hand motion. For example, during $c, \rightarrow, d$, the left hand of the right demonstrator stiffens to push and then becomes compliant to pull back.

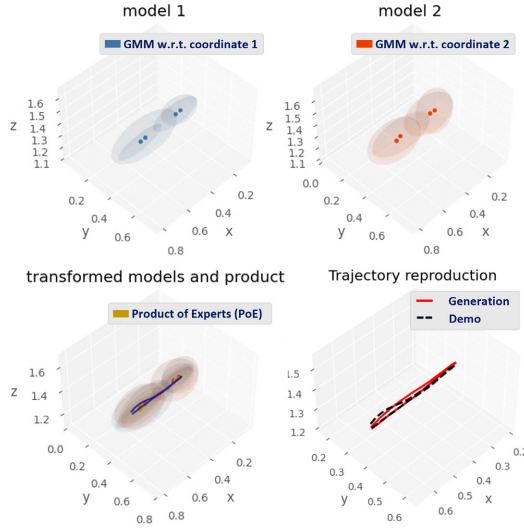


Fig. 4. One example of learning results for motion representation and generation by TP-GMM method is illustrated with respect to the unimanual human-human straight line pulling task.

with $S_K(t)$, $S_D(t)$ I stay non-negative and α, β positive constant, ϵ is non-negative. As a result, the constraint (13) can be satisfied both in the ‘Yin’ and ‘Yang’ phase with the proof (16a) and (16b), respectively.

C. Motion Representation and Trajectory Planning

To learn from human demonstrations, human motions are modeled using TP-GMM and mapped to the robot frame by adjusting task parameters. TP-GMM captures demonstrations from multiple reference frames and employs GMM to group points from the demonstrations into several Gaussian distributions for each frame $j \in [1, P]$. A motion relevant to the task parameters is then reproduced through the product of experts (PoE) [10]. Task parameters refer to factors such as the pose of target objects or the environmental state, depending on the task requirements. Unlike GMM, TP-GMM is used in this work to generate cyclic motions from non-sequential demonstrations. Additional details about the

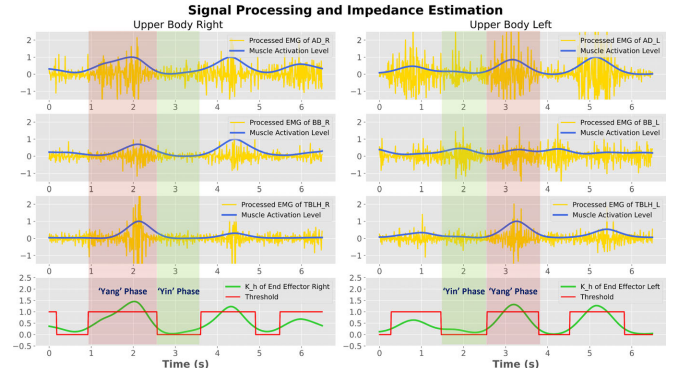


Fig. 5. One example of EMG signal processing and impedance estimation result from human-human demonstrations with three repetitions of flat-circle Tai Chi pushing hands.

approach are available in our prior research [18].

V. EXPERIMENTS

In this section, the software platform used in the experiments is briefly introduced. Then, the trajectory and impedance learning part including data collection, impedance generation, and trajectory learning performance is given. Finally, the real human-robot Tai Chi pulling/pushing hands are presented. The comparative experiments of the proposed variable impedance controller and a fixed impedance controller (FIC) show the advantages of our method to some extent.

A. Experimental Platform

As shown in Fig. 2, we use the CURI robot as the collaborative robot which is introduced in Section II. In this paper, only the robotic manipulators and the torso are applied for Human-Robot Collaboration. For the software development, a whole-body Cartesian impedance control interface is implemented on the basis of the Robot Operating System (ROS). By giving the commanded stiffness and damping matrices with an update frequency of $10Hz$ as well

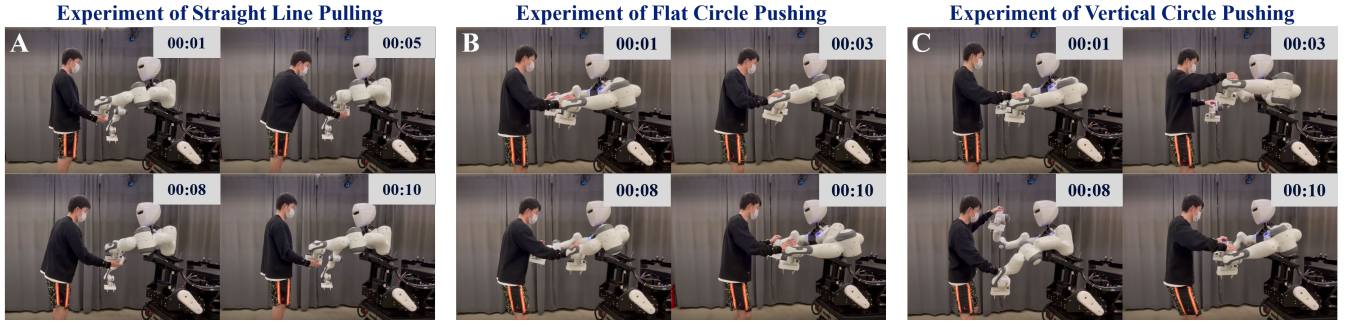


Fig. 6. The snapshots of human-robot confrontational Tai Chi pulling/pushing hands including unimanual straight line pulling (A), bimanual flat circle pushing (B), and bimanual vertical circular pushing (C).

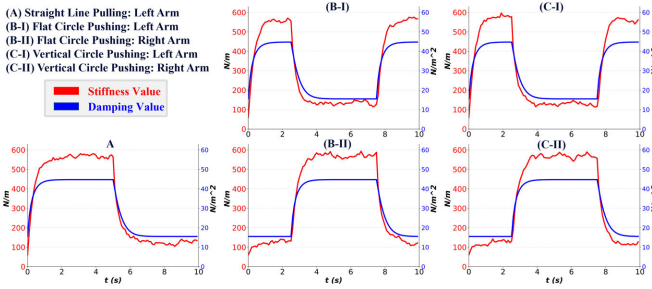


Fig. 7. Variations of stiffness and damping terms of CURI end effectors during the HRI with three Tai Chi pulling/pushing hands tasks.

as the desired trajectory of the end effector, the controller will output the commanded torques of each joint for the robotic manipulators and calculate joint velocities for torso joints to generate expected motions with a frequency of $1000Hz$. Note that the gravity compensation was redesigned due to the unique installations of the dual arms.

B. Trajectory and Impedance Generation

1) *Data Collection*: During data collection, the confrontational Tai Chi pulling/pushing hands motions were recorded using an Xsens suit and EMG sensors, ensuring high dynamic accuracy. The demonstrator's motion, to be imitated by the robot, was captured by IMU sensors on the Xsens suit (attached to the left and right hands) at $60Hz$ (Fig. 3). These data were used for offline learning to generate robot motion representations. Additionally, EMG sensors were placed on the Anterior Deltoid (AD), Biceps Brachii (BB), and Triceps Brachii Lateral Head (TBLH) of both upper body sides to record muscle activity (Fig. 3). The Delsys Trigno Wireless System acquired EMG signals at $2000Hz$. To estimate impedance, we used BoB, a biomechanical modeling software, which integrates Xsens motion data to analyze muscle forces and assign weights to each muscle based on their contribution during the 'Yang' phase. Muscles with significant contributions were assigned higher weights.

2) *Impedance Profile Generation*: The raw EMG signals were processed using a fourth-order $100Hz$ high-pass Butterworth filter and detrending. The cleaned data were then rectified, smoothed at $250Hz$, and normalized to muscle voluntary contraction (MVC) levels. Muscle activation levels and thresholds were calculated to generate impedance

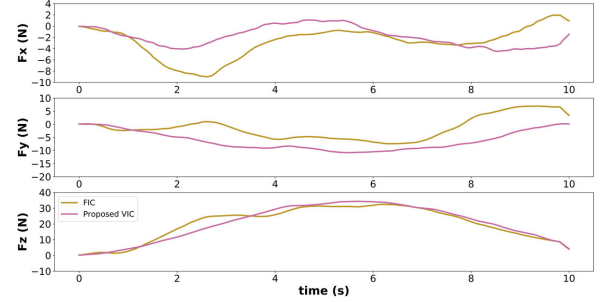


Fig. 8. The comparison of interactive force between FIC with brown line and our proposed VIC method with purple line during the unimanual Tai Chi pulling task.

profiles, which were used to define the variable impedance controller (Section IV-B). Experimentally determined parameters ($\lambda_1 = 20$, $\lambda_2 = 0.15$, $\lambda_3 = 0.25$) guided the profile generation. Fig. 5 illustrates the process, where EMG signals from the demonstrator's upper body (AD: 0.6, BB: 0.3, TBLH: 0.1 weights) were converted into impedance profiles for CURI's arms. The 'Yin' and 'Yang' phases were distinguished based on these profiles and used to design the proposed controller.

3) *Motion Generation and Trajectory Planning*: As shown in Fig. 4, we applied two frames of the human demonstration in GMM to represent the movement. The result of the representation with regard to the PoE was illustrated for the unimanual straight line pulling task in this example. Furthermore, by extending to TP-GMM by adding the task parameter, the motion reproduction was achieved with an end-to-end trajectory and was shown in the fourth graph of Fig. 4.

The collection and preprocessing of the multimodal demonstration data such as IMU and EMG signals, and the implementation of the TP-GMM methods depend on the Rofunc package, a full-process python package for robot learning from demonstration published by our lab [19].

C. Human-Robot Pulling/Pushing Hands

In real world experiment (as illustrated in Fig. 1), the CURI robot was equipped with an Optitrack marker behind its torso in order to obtain the poses of two end effectors in the world coordinate system. Meanwhile, the other two markers were mounted on both hands of the human collaborator. These data were transferred into a rosbag format through

data streaming. Hence, the spatial displacement can be measured from time to time. The movements of CURI were transformed into the robot coordinate by mapping between the marker and the end effectors. The execution of whole-body motion was based on the reference trajectories of each end effector which were learned offline from demonstration via TP-GMM method. Note that the execution time of each repetition is enlarged to 10s in our experiments. The tuning parameters of VIC α and β are set to be 0.05 and the scale index γ is 0.5 in this experiment. K_1 and K_2 are 60 and 500N/m while D_1 and D_2 are 15.5 and 44.7N/m². The predefined safe spatial distance is set to be 0.05m.

Fig. 6 shows snapshots of three human-robot Tai Chi tasks: unimanual straight-line pulling (A), bimanual vertical-circle pushing (B), and bimanual flat-circle pushing (C). Each motion cycle lasted 10s. In the unimanual task, CURI exhibited high impedance to pull the human in the first 5s ('Yang' phase) and passively returned in the next 5s ('Yin' phase). Similarly, during bimanual tasks, CURI's arms alternated between 'Yin' and 'Yang' phases, mimicking the demonstrator. Fig. 7 illustrates the stiffness and damping variations of CURI's end effectors during these tasks. For unimanual pulling, the left arm showed high impedance in the 'Yang' phase and low impedance in the 'Yin' phase. In bimanual tasks, the arms exhibited complementary impedance profiles, with one arm transitioning from 'Yang' to 'Yin' while the other performed the opposite. Small fluctuations in stiffness values indicate stable interaction distances between the human and robot.

To validate the proposed method, we compared it with a FIC. Fig. 8 shows the interactive force between CURI's end effector and the human hand during unimanual pulling. The proposed method resulted in smooth, time-varying force changes, while the FIC caused abrupt force mutations. This demonstrates that our approach enables more natural and stable physical interaction.

VI. CONCLUSIONS

In this paper, we apply a novel humanoid mobile robot CURI to a human-robot collaboration task that performs the Tai Chi pulling/pushing hands with a human partner. We propose a whole-body impedance controller for this unique humanoid robot and put forward an adaptive impedance control method based on the human-human demonstration as well as the spatial displacement to interact with humans in a proper way. Besides, a learning model TP-GMM is proposed for 3D motion representation and generation based on the demonstration. The real-world platform we built successfully achieved the three collaborative Tai Chi pulling/pushing hands task and will be a foundation of future research in HRC tasks. The overall framework we propose to let the cobots learn collaborative skills from human-human demonstration can be extended to other tasks with variable physical behaviors and repetitive actions. For instance, the collaborative carrying objects in logistic scenarios and the assistive dressing clothes in domestic scenarios are feasible applications to achieve.

REFERENCES

- [1] F. Chen, K. Sekiyama, F. Cannella, and T. Fukuda, "Optimal subtask allocation for human and robot collaboration within hybrid assembly system," *IEEE Transactions on Automation Science and Engineering*, vol. 11, no. 4, pp. 1065–1075, 2013.
- [2] B. Nemec, N. Likar, A. Gams, and A. Ude, "Bimanual human robot cooperation with adaptive stiffness control," in *2016 IEEE-RAS 16th International Conference on Humanoid Robots (Humanoids)*, pp. 607–613, IEEE, 2016.
- [3] L. Roza, S. Calinon, D. G. Caldwell, P. Jimenez, and C. Torras, "Learning physical collaborative robot behaviors from human demonstrations," *IEEE Transactions on Robotics*, vol. 32, no. 3, pp. 513–527, 2016.
- [4] L. Pernel, N. Tsagarakis, D. Caldwell, and A. Ajoudani, "Adaptation of robot physical behaviour to human fatigue in human-robot co-manipulation," in *2016 IEEE-RAS 16th International Conference on Humanoid Robots (Humanoids)*, pp. 489–494, IEEE, 2016.
- [5] T. Asfour, M. Waechter, L. Kaul, S. Rader, P. Weiner, S. Ottenhaus, R. Grimm, Y. Zhou, M. Grotz, and F. Paus, "Amar-6: A high-performance humanoid for human-robot collaboration in real-world scenarios," *IEEE Robotics & Automation Magazine*, vol. 26, no. 4, pp. 108–121, 2019.
- [6] M. Sharifi, A. Zakerimanes, J. K. Mehr, A. Torabi, V. K. Mushahwar, and M. Tavakoli, "Impedance variation and learning strategies in human-robot interaction," *IEEE Transactions on Cybernetics*, 2021.
- [7] L. M. Hiatt, C. Narber, E. Bekele, S. S. Khemlani, and J. G. Trafton, "Human modeling for human-robot collaboration," *The International Journal of Robotics Research*, vol. 36, no. 5-7, pp. 580–596, 2017.
- [8] A. Dietrich, *Whole-body impedance control of wheeled humanoid robots*, vol. 116. Springer, 2016.
- [9] C. Ott, *Cartesian impedance control of redundant and flexible-joint robots*. Springer, 2008.
- [10] S. Calinon, "A tutorial on task-parameterized movement learning and retrieval," *Intelligent service robotics*, vol. 9, no. 1, pp. 1–29, 2016.
- [11] O. Khatib, "A unified approach for motion and force control of robot manipulators: The operational space formulation," *IEEE Journal on Robotics and Automation*, vol. 3, no. 1, pp. 43–53, 1987.
- [12] B. Davis, *The taijiquan classics: An annotated translation*. North Atlantic Books, 2004.
- [13] K. Wang, X. Zhang, and H. Zhao, "Biomechanical measurement and analysis of tai chi push-hand," in *2016 IEEE International Conference on Mechatronics and Automation*, pp. 541–545, IEEE, 2016.
- [14] T. Fang, "Yin yang: A new perspective on culture," *Management and organization Review*, vol. 8, no. 1, pp. 25–50, 2012.
- [15] A. Ajoudani, S. B. Godfrey, M. Bianchi, M. G. Catalano, G. Grioli, N. Tsagarakis, and A. Bicchi, "Exploring teleimpedance and tactile feedback for intuitive control of the pisa/iit soft-hand," *IEEE transactions on haptics*, vol. 7, no. 2, pp. 203–215, 2014.
- [16] K. Kronander and A. Billard, "Stability considerations for variable impedance control," *IEEE Transactions on Robotics*, vol. 32, no. 5, pp. 1298–1305, 2016.
- [17] M. Bednarczyk, H. Omran, and B. Bayle, "Passivity filter for variable impedance control," in *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 7159–7164, IEEE, 2020.
- [18] C. Li, X. Wu, T. Teng, S. Calinon, and F. Chen, "Towards robo-coach: Robot interactive stiffness/position adaptation for human strength and conditioning training," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 860–866, IEEE, 2024.
- [19] J. Liu, Z. Dong, C. Li, Z. Li, M. Yu, D. Delehelle, and F. Chen, "Rofunc: The full process python package for robot learning from demonstration and robot manipulation," *Zenodo*, <https://github.com/Skylark0924/Rofunc>, 2023.